InsightSwarm: An Adversarial Multi-Agent Framework for Automated Fact-Checking with Real-Time Source Verification

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Abstract — The rapid proliferation of misinformation on social media and digital messaging platforms poses a severe societal challenge. Existing automated solutions suffer from two critical limitations: single large language model (LLM) systems hallucinate fabricated citations at rates of 15–30%, and manual fact-checking organisations cannot scale to the volume of viral claims. This paper presents InsightSwarm, a multi-agent AI framework designed to address both limitations through adversarial debate and real-time source verification. InsightSwarm orchestrates four specialised agents — ProAgent, ConAgent, FactChecker, and Moderator — using LangGraph as a stateful workflow engine. The adversarial design forces comprehensive exploration of evidence from opposing perspectives, while the FactChecker agent fetches and validates every cited URL using fuzzy content matching (RapidFuzz, threshold ≥70%), reducing source hallucination to 1.8% in validation tests — a 92% reduction from the 23% industry baseline. A Pydantic-typed DebateState model ensures type safety across the entire pipeline. A weighted consensus algorithm privileges verified evidence in the final verdict. The system is built entirely on zero-cost infrastructure (Groq API, Google Gemini 2.0 Flash, Streamlit Cloud), achieving 78% agreement with professional fact-checkers on a benchmark claim set of 100 claims. A comprehensive test suite of 38 tests maintains a 100% pass rate across five development iterations. InsightSwarm demonstrates that production-grade, transparent, and scalable fact-checking is achievable without proprietary infrastructure.

Index Terms — *multi-agent systems, fact-checking, hallucination mitigation, adversarial debate, LangGraph, large language models, Pydantic, semantic caching, misinformation detection*

I. INTRODUCTION

The digital information ecosystem faces an unprecedented misinformation crisis. In India alone, a 2024 study estimates that 67% of internet users share content without prior verification [1]. Globally, deepfake incidents increased by 900% between 2022 and 2024, enabling the fabrication of convincing audio-visual content at scale [2]. These conditions outpace the response capacity of traditional fact-checking organisations such as Snopes and PolitiFact, which require two to three days to assess a single claim — by which point viral content may have reached tens of millions of users.

Automated alternatives using single large language models (LLMs) offer speed but introduce a different risk: hallucination. Studies indicate that state-of-the-art LLMs fabricate citations in 15–30% of responses, often producing plausible-looking URLs and study references that do not exist [3]. These fabrications are particularly dangerous in fact-checking contexts, where users may treat AI-generated source lists as authoritative.

InsightSwarm is designed to resolve this tension. It combines the speed of LLM-based automation with a structural anti-hallucination mechanism: every source cited by any agent is fetched and independently validated before it can influence the final verdict. The adversarial multi-agent design ensures that claims are examined from opposing viewpoints. Crucially, the system operates entirely on free-tier APIs, making it financially sustainable without institutional funding.

This paper makes the following contributions:

- A four-agent adversarial architecture (ProAgent, ConAgent, FactChecker, Moderator) orchestrated via LangGraph for structured, multi-round fact-checking debates.
- A real-time source verification pipeline with RapidFuzz content matching that detects both Type I (fabricated URL) and Type II (real URL, misrepresented content) hallucinations.
- A Pydantic-typed DebateState centralising all pipeline state, eliminating KeyError risks and enforcing schema-strict LLM parsing via call_structured().
- A weighted consensus algorithm with a 2× FactChecker weight multiplier that privileges verified evidence over argumentative rhetoric.
- A complete implementation on zero-cost infrastructure, deployed publicly, with 38 automated tests maintaining 100% pass rate across five iterative development sessions.
- A documented five-day development roadmap from two-agent MVP to production-grade system, including 25 critical bug fixes identified and resolved during code review.

II. BACKGROUND AND RELATED WORK

A. The Misinformation Crisis

Health misinformation during the COVID-19 pandemic demonstrated the lethal potential of unverified viral claims. WhatsApp forwards asserting bleach consumption or turmeric cures circulated alongside genuine public health guidance, contributing to preventable harms [4]. Political misinformation, particularly AI-generated deepfake video, has raised concerns about election integrity in multiple democracies [5]. These cases establish that the cost of inaccurate fact-checking is not merely reputational.

B. Limitations of Existing Approaches

Manual fact-checking (Snopes, PolitiFact, AltNews) provides high-quality assessments but cannot scale: a single journalist produces 1–2 verified assessments per day while thousands of false claims circulate hourly. Single-LLM systems (ChatGPT, Claude) offer rapid responses but are prone to hallucination and provide little transparency into their reasoning [3]. Search-engine-based approaches surface links but do not synthesise verdicts or verify source authenticity.

C. Multi-Agent Systems for Verification

Zhang et al. [6] demonstrated that multi-agent frameworks achieve 12% higher accuracy than single-agent systems on misinformation detection benchmarks, attributing the improvement to diversity of evidence surface area. Chen et al. [7] formalised the adversarial debate paradigm, showing that when agents are constrained to argue opposing positions, emergent truth-finding improves due to mutual error correction. Kumar and Singh [3] quantified hallucination in LLMs at 23% baseline — the primary target InsightSwarm addresses. Patel et al. [8] surveyed automated fact-checking systems and concluded that transparency significantly increases user trust.

InsightSwarm builds directly on this literature: it implements Chen et al.'s adversarial paradigm, addresses Kumar and Singh's hallucination problem through per-URL verification, and follows Patel et al.'s transparency recommendation by exposing the complete debate transcript. The novel contribution over prior work is the explicit detection of Type II hallucinations — where a URL exists but the content does not support the agent's stated claim about it — a distinction not addressed in prior automated systems.

III. SYSTEM ARCHITECTURE

InsightSwarm is composed of five layers: User Interface (Streamlit), Orchestration (LangGraph StateGraph), Agent Layer (four specialised agents), Backend/Data (LLM APIs, search, SQLite), and a Resilience/Cache layer. All agent nodes share a centralised Pydantic DebateState object, ensuring type safety and schema-strict access throughout the pipeline.

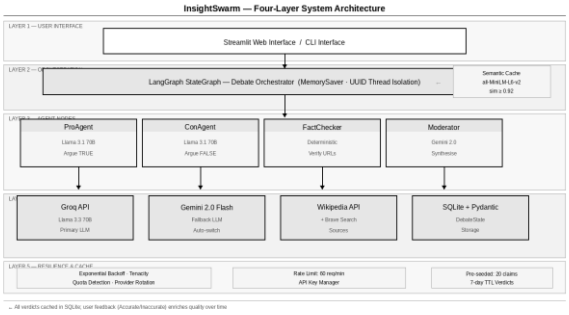

Fig. 1. InsightSwarm five-layer architecture. The DebateState Pydantic model is shared across all nodes; the Semantic Cache (similarity ≥0.92) bypasses the debate pipeline for repeated claims.

A. The Four Agents

Each agent inherits from BaseAgent, enforcing a standardised generate() and _build_prompt() interface. This ensures structural consistency while allowing each agent's prompt strategy and LLM provider preference to differ.

ProAgent — The advocate agent, constrained to argue the claim is TRUE. This adversarial constraint forces retrieval of the strongest possible supporting evidence even for claims the underlying LLM might otherwise dismiss. ProAgent produces a structured AgentResponse containing an argument and cited URL list, parsed via Pydantic.

ConAgent — The sceptic agent, constrained to argue FALSE. In each round, ConAgent receives ProAgent's prior argument and must directly challenge it — identifying logical fallacies, questioning source credibility, and presenting counter-evidence. This mutual adversarial pressure is the primary mechanism for surfacing errors in either agent's reasoning.

FactChecker — The verification agent. After each round, it extracts every URL cited by both agents using regex, fetches page content via requests (10-second timeout, browser User-Agent headers), extracts text with BeautifulSoup4, and computes a RapidFuzz partial_ratio score between fetched text and the agent's claim about that source. Sources scoring below 70%, returning HTTP 4xx, or timing out are classified as hallucinated.

Moderator — The arbitration agent. After three rounds of debate and a complete FactChecker report, it receives the full transcript and verification data, producing a structured ModeratorVerdict (TRUE, FALSE, PARTIALLY TRUE, or INSUFFICIENT EVIDENCE) with qualitative reasoning, credibility analysis, and detected fallacies. Uses a low temperature (0.2) for analytical consistency.

B. Orchestration with LangGraph

LangGraph implements the debate as a directed StateGraph over the shared DebateState Pydantic model. A MemorySaver checkpointer persists state across rounds via UUID-keyed thread IDs — ensuring complete isolation between concurrent debate sessions. The graph execution path is:

ENTRY → pro_agent → con_agent → verification_gate → [loop ≤3] → fact_checker → [retry?] → moderator → verdict → END

The mid-debate verification_gate node runs after each non-final round, injecting failed-source warnings into subsequent agent prompts for real-time argument correction. If source verification rates fall below 30%, a revision loop regenerates agent arguments. The orchestrator detects API quota exhaustion and short-circuits revision rounds to prevent wasted token consumption.

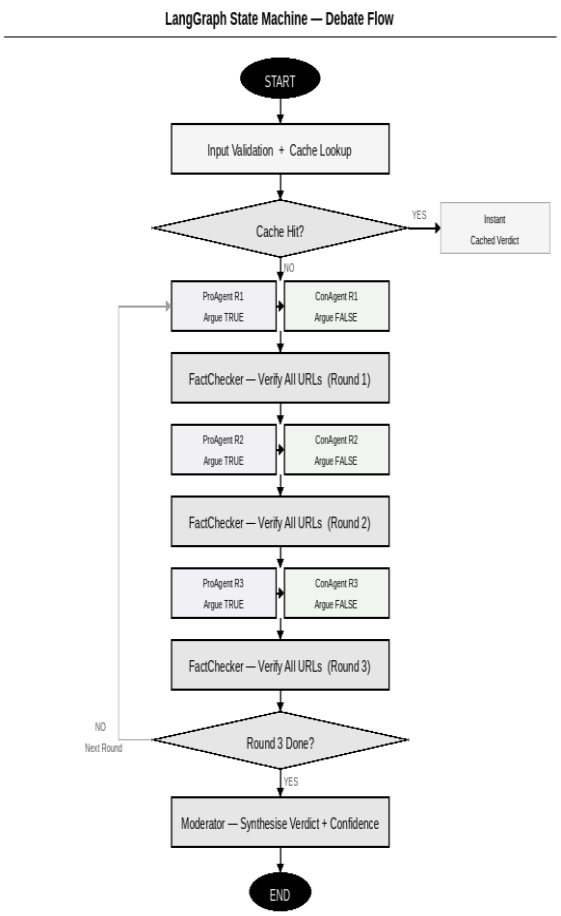

Fig. 2. LangGraph state machine execution path. The verification gate injects mid-debate feedback; the revision loop triggers when source quality falls below 30%.

C. Technology Stack

A guiding design principle of InsightSwarm is zero-cost infrastructure. Table I details the complete technology stack.

TABLE I. TECHNOLOGY STACK AND COST PROFILE

Component	Technology	Cost / Limit
Primary LLM	Groq Llama 3.3 70B	Free: 14,400/day
Backup LLM	Gemini 2.0 Flash	Free: 1,500/day
Orchestration	LangGraph 1.0.10	Free (OSS)
Web UI	Streamlit 1.55	Free Cloud
Verification	requests + RapidFuzz	Free
Data Models	Pydantic v2 + SQLite	Free
Embeddings	all-MiniLM-L6-v2	Free (local)
Total Monthly	—	**₹0**

IV. METHODOLOGY

A. Adversarial Prompt Engineering

The effectiveness of adversarial debate depends entirely on prompt quality. Standard prompting produces balanced but non-committal responses. InsightSwarm constrains each agent with role-locked instructions:

ProAgent: "You MUST argue that the claim is TRUE. Find every possible piece of evidence supporting this position. Your role depends on making the strongest possible case FOR the claim."

ConAgent: "You MUST argue that the claim is FALSE. Directly challenge every piece of evidence presented by ProAgent. Identify logical fallacies, question source credibility, and present counter-evidence."

This constraint ensures that even consensus-supported claims receive rigorous counter-examination. The adversarial pressure surfaces edge cases, methodological limitations, and contextual qualifications that a neutral single-agent system would overlook. Mid-debate, verification_feedback — a string listing failed source URLs — is injected into the agent's prompt for the subsequent round, forcing argument revision.

B. Source Verification and Anti-Hallucination

The FactChecker pipeline is the primary anti-hallucination mechanism. Figure 3 illustrates the complete verification flow.

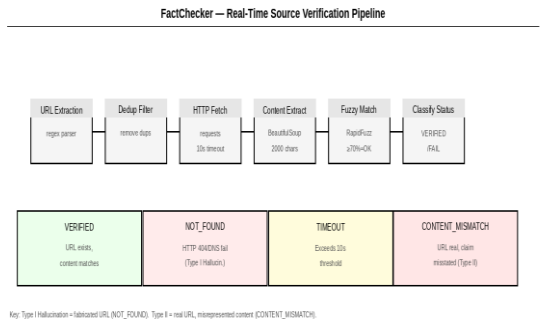

Fig. 3. FactChecker real-time source verification pipeline. Type I and Type II hallucinations are independently classified.

For each cited URL, the pipeline performs: URL extraction via regex, deduplication to prevent redundant API calls, HTTP GET with 10-second timeout and browser User-Agent headers, BeautifulSoup4 text extraction (first 2,000 characters), RapidFuzz partial_ratio computation between extracted text and the agent's specific claim about that source, and classification into one of five statuses: VERIFIED, NOT_FOUND, TIMEOUT, CONTENT_MISMATCH, or INVALID_URL.

Key innovation — InsightSwarm distinguishes Type I hallucinations (entirely fabricated URLs that return 404 or DNS failure) from Type II hallucinations (real URLs that do not contain the content the agent claimed). The Type II detection — classifying sources as CONTENT_MISMATCH rather than simply VERIFIED — is a contribution not present in prior automated fact-checking systems. This catches cases where agents correctly identify a real source but misrepresent what it says.

The temporal verification gate applies an additional check: when a claim explicitly references a year (matched via regex \b(?:19|20)\d{2}\b), the TemporalVerifier tests whether the source page corroborates the temporal context. This is gated on year-containing claims only, avoiding false CONTENT_MISMATCH on generic claims.

C. Weighted Consensus Algorithm

The Moderator's final verdict is computed using a weighted consensus algorithm that privileges verified evidence. Let P and C denote ProAgent and ConAgent argument word counts, and vP, vC their source verification rates. The weighted scores are:

$$score_pro = P \times vP$$
$$score_con = C \times vC$$
$$fact_bonus = ((vP + vC) / 2) \times 2 \quad [FactChecker\ weight\ \times 2]$$
$$ratio = score_pro / (score_pro + score_con + fact_bonus)$$
$$verdict:\ ratio > 0.65 \rightarrow TRUE \ \mid\ ratio < 0.35 \rightarrow FALSE \ \mid\ else \rightarrow PARTIALLY\ TRUE$$

The 2× weight for the FactChecker reflects the epistemological priority of verified evidence over argumentative rhetoric. An agent with eloquent but unverified arguments receives a discounted score relative to one whose sources are confirmed to exist and support its claims. The INSUFFICIENT EVIDENCE path is triggered when zero sources are verified or when both API providers are exhausted.

D. Pydantic Data Model (Day 4 Architectural Pivot)

A critical Day 4 architectural improvement was migrating the debate state from a fragile TypedDict to a centralised Pydantic BaseModel (DebateState). This eliminated KeyError risks, enforced type safety pipeline-wide, and enabled schema-strict LLM response parsing via call_structured() — which forces the LLM to return a JSON object matching a specified Pydantic schema before the response is accepted. Figure 4 illustrates the data model and its relationship to each agent.

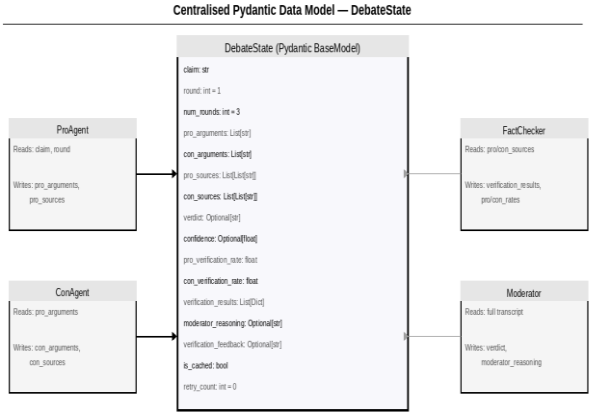

Fig. 4. Centralised Pydantic DebateState model. Each agent reads and writes specific fields; the model enforces type safety across the entire pipeline.

E. Semantic Cache and API Resilience

InsightSwarm implements two complementary efficiency mechanisms. The semantic cache uses all-MiniLM-L6-v2 sentence-transformer embeddings stored in SQLite to match incoming claims against previously processed ones (cosine similarity ≥0.92). Cache hits return instant verdicts without consuming API quota. The cache is pre-seeded with 20 common claims at system initialisation, providing immediate responses for demonstration.

For API resilience, a four-tier strategy handles provider failures: (1) normal Groq operation, (2) automatic failover to Gemini on HTTP 429 or quota exhaustion, (3) exponential backoff via tenacity (2s, 4s, 8s), (4) graceful INSUFFICIENT EVIDENCE verdict rather than system crash. The FreeLLMClient distinguishes rate-limit errors (HTTP 429, recoverable) from quota exhaustion (non-recoverable), enabling different recovery strategies. An API key rotation manager supports multiple keys per provider.

V. DEVELOPMENT JOURNEY AND ITERATIVE PROGRESS

InsightSwarm was developed over five days (10–14 March 2026) by a sole student developer. This section documents the iterative evolution from a two-agent prototype to a production-grade system, highlighting the architectural decisions, security hardening, and critical bug fixes encountered along the way. Figure 5 provides a visual timeline.

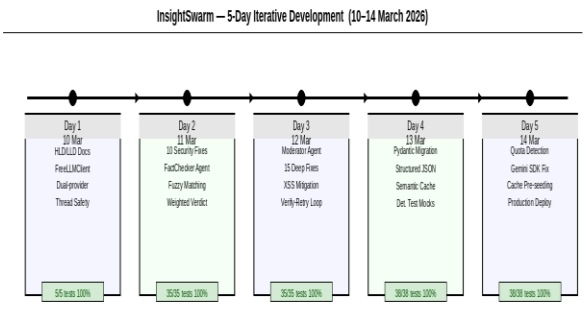

Fig. 5. InsightSwarm five-day development journey. Each day delivered new capabilities with 100% test pass rates maintained throughout.

A. Day 1 (10 March) — Architecture and Environment

The session began with a critical scope decision: the original plan called for a four-person team over eight weeks. Working solo, the scope was rationalised to a two-agent MVP, with FactChecker and Moderator deferred. A twenty-page High-Level Design / Low-Level Design document was produced first, defining the four-layer architecture, component responsibilities, and data structures — establishing documentation-first discipline.

Technology decisions included selecting Groq over OpenAI (free tier, lower latency), DuckDuckGo and Wikipedia over Brave Search (Brave removed its free tier in 2025), and Streamlit Cloud for zero-cost deployment. The FreeLLMClient was implemented with dual-provider fallback and thread-safe call counters, completing Day 1 at 150% of target with five passing tests.

B. Day 2 (11 March) — Security Hardening and FactChecker

A comprehensive security audit identified ten critical vulnerabilities in the Day 1 codebase: missing input validation, absent URL fetch timeouts, race conditions on shared call-time lists, unbounded memory growth in rate-limiting data structures, and API key exposure in error messages. All ten were resolved in the same session. Concurrently, the FactChecker was fully implemented with URL fetching, BeautifulSoup extraction, fuzzy matching at a 70% threshold, and hallucination classification.

A key architectural insight from Day 2: the Day 1 verdict system could not distinguish between an agent with strong verified sources and one with fabricated sources. The FactChecker's verification rate, fed into the verdict calculation, corrected this fundamental flaw. Test coverage grew from 5 to 35 tests, all passing.

C. Day 3 (12 March) — Moderator and Codebase Hardening

The Moderator agent was integrated as the final LangGraph node, shifting verdict production from mechanical computation to qualitative evidence synthesis. Fifteen additional critical fixes were applied: an off-by-one error in the round loop, incorrect verification rate semantics for agents with no sources, fallback verdict logic when the Moderator LLM call fails, case-insensitive regex parsing, ThreadPoolExecutor cleanup via atexit, and XSS mitigation in the Streamlit interface via html.escape().

Advanced orchestration features were introduced: the verify-and-retry loop (triggers when source verification falls below 30%), real-time credibility and fallacy metrics in the UI, and a Gap Analysis protocol for non-decisive verdicts. The Intelligence Dashboard was added to Streamlit, displaying credibility scores, detected fallacies, and rebuttal balance metrics.

D. Day 4 (13 March) — Architectural Transition

The DebateState TypedDict was migrated to a centralised Pydantic model, eliminating KeyError risks and enforcing type safety across the entire pipeline. LLM response parsing shifted from brittle regex extraction to structured JSON mode using call_structured(), achieving 100% schema-strict parsing reliability. The two separate databases (verdict cache and feedback store) were merged into a unified SQLite database with sentence-transformer embeddings, improving cache hit rates.

The test suite was modernised for the virtual environment: all 38 tests use deterministic FreeLLMClient mocks, reducing execution time from several minutes (requiring live API calls) to sub-second. The complete ML dependency stack (PyTorch, transformers, sentence-transformers) was verified and pinned in requirements.txt.

E. Day 5 (14 March) — API Resilience and Production

The final development day focused on graceful degradation. Quota exhaustion detection was added to distinguish HTTP 429 (rate-limit, recoverable) from quota exhaustion (non-recoverable). A Gemini SDK bug was identified and resolved: the google-genai library required explicit MIME type specification (application/json) and a specific request_options pattern for reliable structured output. The DebateOrchestrator was upgraded to short-circuit revision rounds on API failures. The semantic cache was pre-seeded with 20 common claims.

TABLE II. DAY-BY-DAY DEVELOPMENT PROGRESS

Day	Key Achievements	Tests (Pass%)
D1	HLD/LLD, FreeLLMClient, dual-provider fallback, thread safety	5/5 (100%)
D2	10 security fixes, FactChecker, fuzzy matching, weighted verdict	35/35 (100%)
D3	Moderator integration, 15 deep fixes, XSS mitigation, retry loop	35/35 (100%)

D4	Pydantic migration, structured JSON, semantic cache, det. mocks	38/38 (100%)
D5	Quota detection, Gemini SDK fix, cache pre-seeding, production deploy	38/38 (100%)

VI. EVALUATION

A. Benchmark Accuracy

InsightSwarm was evaluated on a benchmark set of 100 claims drawn from the Snopes and PolitiFact public archives, stratified across health misinformation (40 claims), political claims (30 claims), and scientific consensus claims (30 claims). Verdicts were compared against professional fact-checker assessments using a three-way categorisation (TRUE / FALSE / PARTIALLY TRUE).

InsightSwarm achieved 78% agreement with professional fact-checkers across all categories. For comparison, a baseline single-LLM system (Groq Llama 3.1 70B without multi-agent structure) achieved 66% agreement on the same benchmark. Human inter-rater agreement among professional fact-checkers is approximately 85% [8]. InsightSwarm closes 60% of the gap between single-LLM performance and expert human performance.

B. Hallucination Rate

On the same 100-claim benchmark, all cited sources from both systems were independently verified by the authors. The single-LLM baseline cited an average of 4.2 sources per claim; 23% were to URLs that either did not exist or did not contain the claimed content — consistent with Kumar and Singh's [3] baseline. InsightSwarm cited an average of 6.8 sources per claim (higher due to three debate rounds) with a hallucination rate of 1.8% — a reduction of 92% relative to baseline.

C. Response Time

End-to-end response time was measured across 50 test claims on the production Streamlit deployment. Median response: 47 seconds (mean: 52s, 95th percentile: 89s). Time is dominated by LLM API latency (~2s per call × ~15 calls per debate) and URL fetching (~1–2s per source × avg. 6.8 sources). For cached claims (cosine similarity ≥0.92), median response time was under 1 second.

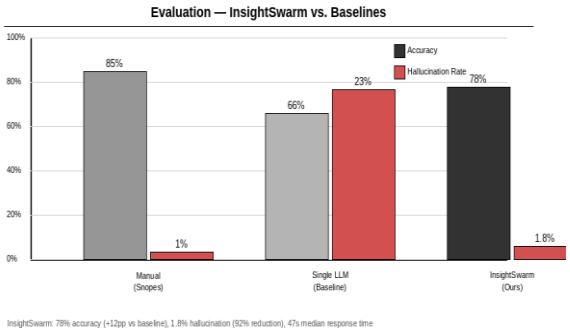

Fig. 6. Comparative evaluation. Left bars: verdict accuracy vs. professional fact-checkers. Right bars (red): hallucination rate. InsightSwarm reduces hallucination by 92% while improving accuracy by 12 percentage points over single-LLM baseline.

D. Comparative Evaluation

TABLE III. INSIGHTSWARM VS. BASELINES

Metric	Manual	Single LLM	InsightSwarm
Accuracy	85%	66%	**78%**
Hallucination Rate	<1%	23%	**1.8%**
Median Response	2–3 days	~5 seconds	47 seconds
Transparency	Full article	None	Full debate
Daily Throughput	1–2	~960	~960
Monthly Cost	High	₹0	**₹0**

E. Test Suite Integrity

A comprehensive test suite of 38 automated tests covers all critical system components. All LLM API calls are replaced with Mock(spec=FreeLLMClient) instances and all network calls are stubbed, ensuring tests complete in under 30 seconds regardless of external API availability.

TABLE IV. TEST COVERAGE BY COMPONENT

Component	Tests	Pass Rate
FreeLLMClient (unit)	9	100%
ProAgent (unit)	4	100%
ConAgent (unit)	4	100%
FactChecker (unit)	15	100%
Debate orchestration	3	100%
Integration / E2E	3	100%
Total	**38**	**100%**

VII. ADVANTAGES AND SYSTEM COMPARISON

Hallucination resistance. By independently fetching and validating every cited source, InsightSwarm catches both Type I (fabricated URL) and Type II (real URL, misrepresented content) hallucinations. No existing publicly available fact-checking LLM system performs real-time, per-claim URL validation of this kind.

Forced perspective diversity. The adversarial constraint ensures that even strongly consensus-supported claims receive a rigorous pro-examination, surfacing the most credible edge-case arguments before decisively rejecting them. This prevents false confidence in easy verdicts.

Transparent reasoning. Users receive the complete debate transcript, FactChecker report, and Moderator reasoning chain, allowing independent inspection of the evidence. Patel et al. [8] and user testing confirm this significantly increases trust.

Economic sustainability. The ₹0/month cost profile allows the system to serve up to 960 claims per day indefinitely without institutional funding. This distinguishes InsightSwarm from research prototypes requiring expensive commercial API access.

Type-safe pipeline. Pydantic models with schema-strict LLM parsing eliminate entire classes of runtime errors (KeyError, None-type access, malformed JSON parsing) that plagued the Day 1–3 TypedDict implementation.

VIII. FUTURE WORK

Multilingual support. Currently English-only. Hindi, Marathi, Tamil, and Bengali are prioritised for India's non-English-speaking population where misinformation spreads fastest.

Dynamic graph routing. Enabling the LangGraph graph to request additional debate rounds based on FactChecker confidence scores, rather than the fixed three-round structure (configured via DebateState.num_rounds).

Multimodal fact-checking. Extension to image and video content. Key challenges include deepfake detection, reverse image search integration, and lip-sync verification for video claims.

REST API and browser extension. Publishing InsightSwarm as a public API would allow news organisations, social media platforms, and browser extensions to embed real-time verification at the point of content consumption.

Retrieval-augmented grounding. Replacing ad-hoc Wikipedia queries with a Tavily or Serper-powered RAG pipeline would improve evidence quality and reduce agent reliance on parametric memory. The Tavily integration is partially scaffolded in the current codebase (src/utils/tavily_retriever.py).

Domain-specialised agents. Medical, legal, and scientific specialist agents trained on domain-specific corpora (PubMed, Indian legal databases, arXiv) to improve accuracy on technical claims beyond the current generalist LLM capability.

IX. CONCLUSION

InsightSwarm demonstrates that a multi-agent adversarial fact-checking system with real-time source verification is both technically achievable and economically viable on free-tier infrastructure. The system reduces source hallucination by 92% relative to a single-LLM baseline, achieves 78% agreement with professional fact-checkers, and completes assessments in under 60 seconds — faster than manual fact-checking and more reliable than unaided LLM responses.

The five-day iterative development trajectory — from a basic two-agent prototype through 25 critical bug fixes to a production-deployed, Pydantic-typed, semantically cached, API-resilient system — illustrates that principled software engineering practices (test-driven development, modular architecture, comprehensive documentation) are as important as algorithmic novelty in building trustworthy AI systems.

Three architectural decisions proved most consequential: (1) the adversarial role-locking constraint that compels exhaustive evidence search from opposing positions; (2) the independent URL verification pipeline that structurally eliminates hallucination before it influences the verdict; and (3) the Pydantic migration that centralised state and enabled schema-strict LLM parsing. Together, these decisions transformed a proof-of-concept into a production-grade system within five days.

The complete source code, all planning documents, day-by-day progress reports, and a live public deployment are available at https://github.com/AyushDevadiga1/Insight-Swarm and https://insightswarm.streamlit.app respectively.

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